

SW332 Project – Part 1

In this project, the students will have to implement a web or a mobile application of their choice. To do that they will follow the scrum process.

The project is divided into three parts, each part will be submitted on the due date.

- ✓ Part 1: Team formation and product inception
- ✓ Part 2: First Sprint
- ✓ Part 3: Second Sprint

For each submission, you will have to submit a report and fill in the given templates.

The following table provides the time distribution and the due dates of each step:

Submitting on Blackboard

	Start Date	Submission date
Part 1	W08 October 12, 2025	W09 October 19, 2025
Part 2	W09 October 19, 2025	W11 November 2, 2025
Part 3	W11 November 2, 2025	W13 November 16, 2025

Part 1 of 3: Team formation and product inception

This part includes the processes related to initiation of a project: Create Project Vision, Identify Scrum Master and Stakeholder(s), Form Scrum Team, Develop Epic(s), Create Prioritized Product Backlog, and Conduct Release Planning.

The aim of the SW332 project is to form and work as Scrum Teams to build a real application (Web and/or Mobile). The project starts at the end of Week 8 and ends at Week 13.

The first step of the project is as follows:

- 1) Team creation. The goal is: self-organized team.

Roaa Alhumaidi	445009157
Joud Alqazlan	445009165
Bushra Almughirah	445009189
<u>Sumaiya Balatif</u>	4450094122

- 2) Create the product vision (**Use the Vision Statement Template**)

A: Write the Vision Statement

- **Product Name** Hospital management system
- **For:** Doctor, Patient, Nurse, Administrator, Pharmacist
- **Who** Need one single and efficient system to drive hospital operations, patient data, and communication within a hospital.

- **The** HMS is a comprehensive hospital management system that unifies all departments onto one platform.
- **That** simplifies day-to-day hospital activities, reduces paperwork, improves interdepartmental coordination, and enables better patient care by providing accurate and timely information flow.
- **Unlike** traditional paper or stand-alone systems that cause delays, data loss, and mistakes.
- **Our product** transforms healthcare administration by providing a smart, simple, and fully integrated setting that enhances hospital efficiency, accuracy, and patient satisfaction.

B: Summary of the Product:

The Hospital Management System is an end to end integrated system that is aimed at enhancing hospital efficiency and patient treatment. It consolidates on a daily basis the operations of the hospital from appointment scheduling, patient history, billing, prescriptions, laboratory results, and staff management into one system.

C: Stakeholders:

Internal Stakeholders:

- Hospital Administrators
- Doctors
- Nurses
- Pharmacists: inventory
- Accountants
- IT/Development Team

External Stakeholders:

- Patients
- Insurance Companies
- Health Authorities

D: Team Members (Key Roles):

Product Owner: Sumaiya

Scrum Master: Joud

Development Team Member: Roaa

Development Team Member: Bushra

- 3) Propose a distribution of the scrum roles of your team members **(This distribution will change in each step of the project)**

Product Owner	<u>Sumaiya Balatif</u>	ensures that the hospital system meets the needs of stakeholders by clearly defining the project vision and prioritizing tasks based on value.
Scrum Master	Joud Alqazlan	guides the team in following Scrum practices, removes any obstacles hindering progress, and fosters a collaborative environment to ensure the team stays focused and productive.
Development Team	Roaa Alhumaidi	writes the Product Backlog , prioritizes tasks, focus on delivering functional components and review progress during the Sprint Review .
Development Team	Bushra Almughirah	contributes to designing, developing, and testing the system. They work closely with the team to ensure smooth execution of the sprint and help achieve the set goals.

- 4) Create your initial product backlog **(Use the Product Backlog Template)**. You must have a minimum of 15 items, not detailed. Do not forget that Not all items in a product backlog will be at the same level of detail at the same time.
- Create initial product backlog items as stories **(could be epics or user stories)**,

b. Order your product backlog by priority.

PRODUCT BACKLOG

PRODUCT BACKLOG

ID	AS A ...	I WANT TO ...	SO THAT ...	PRIORITY	Number of SPRINTs	STATUS
1	Administrator	register doctors, nurses, and staff	I can manage user access and system roles	High	1	Complete
2	Patient	register and create an account	I can book appointments and access my medical records	High	1	Complete
3	Patient	book an appointment online	so that I can choose a convenient time with my preferred doctor	High	2	Complete
4	Doctor	view my patient list and appointments	I manage my daily schedule efficiently	High	1	In Progress
5	Doctor	view a patient's full medical record before treatment	so that I can make accurate diagnoses and decisions	High	1	In Progress
6	Patient	enter my insurance information in the hospital application	So that I can check if my insurance is valid and see my coverage amount	High	1	Not Started
7	Lab Technician	upload test results to the system	doctors can access lab data without delay	High	2	Not Started
8	Nurse	update patient's vital signs and daily progress	doctors can monitor patient condition in real life	High	1	Not Started
9	Accountant	generate billing and payment reports	I can track hospital revenue and patient payments	Medium	2	Not Started
10	Doctor	request lab tests and view results digitally	I can make faster medical decisions	Medium	2	Not Started
11	Pharmacist	manage and update available medicines in the inventory	I can ensure that prescriptions are filled correctly	Medium	1	Not Started
12	Administrator	generate reports on hospital performance and staff activity	I can evaluate efficiency and improve management	Medium	2	Not Started
13	Patient	pay medical bills online securely	I can complete my payments easily from home	Medium	2	Not Started
14	Patient	view prescriptions and lab results online	I can stay informed about my health status	Medium	2	Not Started
15	Patient	receive appointment reminders via SMS or email	I won't forget my scheduled visits	Low	1	Not Started
16	Doctor	communicate with other doctors through secure messaging	I can discuss patient cases easily	Low	2	Not Started
TOTAL					24	

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PBIs distribution:

Roaa	PBI-1 to PBI-4
Bushra	PBI-5 to PBI-8
Joud	PBI-9 to PBI-13
Sumaiya	PBI-14 to PBI-16

5) Create the first Sprint Backlog (Use the Sprint Backlog Template).

SPRINT BACKLOG

CURRENT DATE	SPRINT START DATE	TOTAL ASSIGNED HOURS	HOURS USED	HOURS REMAINING
10/16/25	10/16/25	61	43	18
SPRINT GOAL	the team aims to build the core functionalities of the system to ensure smooth workflow			

SPRINT TRACKING

					Start of the Sprint Date								
User Stories and Tasks		Assigned To	Assigned Hours	Hours Used	Available Hours Remaining	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8
User Story 1 Name: as an Administrator I want to register hospital staff so that I can manage user access and system roles			5	5	0	5	0	0	0	0	0	0	0
Task 1.1 Build interface for registering hospital staff	Member's Name: Roaa	3	2	1	2	0	0	0	0	0	0	0	0
Task 1.2 Integrate the system with database	Member's Name: bushra	3	3	0	3	0	0	0	0	0	0	0	0
User Story 2 Name:as a doctor I want to view full medical records of patients before treatment so that I can make accurate diagnoses and decisions			4	2	2	0	2	0	0	0	0	0	0
Task 2.1 design interface for display patients medical records	Member's Name: joud	3	2	1	0	2							
User Story 3 Name: as a patient I want to register and create an account so that I can book appointments and access my medical records			8	6	2	5	1	0	0	0	0	0	0
Task 3.1 develop the functionality of appointment booking	Member's Name :joud	3	3	0	3	0	0	0	0	0	0	0	0
Task 3.2 build interface for patients registration and integrate it with database	Member's Name: Roaa	4	2	2	2	0	0	0	0	0	0	0	0
Task 3.3 design interface to display patients medical record	Member's Name: sumaiya	1	1	0	1	0	0	0	0	0	0	0	0
User Story 4 Name: as a doctor I want to view my patient list and appointments so that I manage my daily schudeles efficiently			6	3	3	0	0	3	0	0	0	0	0
Task 4.1 Develop the functionality to retrieve and display patient list	Member's Name :Sumayia	4	4	0	0	0	2	0	0	0	0	0	0
Task 4.2 implement a search /filter for appointment based on date or patient	Member's Name :bushra	5	2	3	0	0	2	0	0	0	0	0	0
User Story 5 Name: as a patient I want to book appointments online so that I can choose a convenient time with my preferred doctor			12	8	4	0	0	3	3	2	0	0	0
Task 5.1 Design UI for selecting doctor and viewing available slots	Member's Name :Roaa	4	3	1	0	0	3	0	0	0	0	0	0
Task 5.2 Develop backend API to fetch doctors' availability	Member's Name :Sumayia	8	5	3	0	0	0	3	2	0	0	0	0
TOTAL			61	43	18								

<https://pnu.edu->

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6) You should meet every 2 days (at least). (Use the Scrum Meetings Template)

DAILY SCRUM						
Team member	Question	Sunday	Monday	Tuesday	Wednesday	Thursday
Sumaiya	What did you do yesterday?	-	Attend team meeting.	Wrote the first draft of product vision.	Reviewed the vision with team.	Prepares notes for next Scrum meeting.
	What are doing today?	Sprint up plan.	Finalizing project goal.	Sharing feedback with members.	Organizing presentation points.	Summarizing weekly work.
	Is there anything blocking you?	No	No.	No.	No.	No.
Joud	What did you do yesterday?	-	Reviewed user stories.	Created the first sprint Backlog.	Organized tasks by priority.	Updated task status.
	What are doing today?	Following up with team.	Selecting tasks for sprint.	Updating the table.	Checking progress.	Writing summary notes.
	Is there anything blocking you?	No	No.	No.	No.	No.
Roaa	What did you do yesterday?	-	Collected requirements from team.	Added items to Product Backlog.	Reviewed task distribution with Busra.	Updated final list.
	What are doing today?	Organize backlog.	Assigning task.	Tracking updates.	Reviewing team items.	Reviewing work with team.
	Is there anything blocking you?	No	No.	No.	No.	No.
Bushra	What did you do yesterday?	-	Helped Roaa distribute tasks.	Checked task status.	Followed up with team progress.	Added short notes for review.
	What are doing today?	Organizing work.	Updating progress.	Writing documentation.	Reviewing updates with Roaa.	Preparing weekly report.
	Is there anything blocking you?	No.	No.	No.	No.	No.

Rubric

Project part 1 of 3

Project part 1 of 3			
Criteria		Team or Student	Pts checklist
Self-organized team		T	2 P
Distribution of the scrum roles		T	3
Vision Statement File		T	13
Product Backlog		T	15
A	Create initial product backlog items. Write at least 15 PBIs.	T	10
B	Each member has to write at least 3 PBIs.	S	3
C	All product backlog items are expressed as excellent epics and/or user stories. The grade is given to the team if all Backlog rules are ok (DEEP)	T	2
Write the Sprint Backlog for Sprint 1		T	12
Write in the report meetings summary. You should have a meeting at least every 2 days.		T	5
Total Points: 50			

NOTE: The grades distribution might be revised and changed. You will be informed if it happens.